K. D. Polytechnic, Patan

Department of Computer Engineering

Course: Information Security

Course Code: DI05007021

Assignment – 3

1. Explain the concept of Asymmetric Key Encryption with a neat diagram.
2. Differentiate between public key encryption and private key encryption with suitable examples. (Modified)
3. Calculate public and private keys using two prime numbers, p = 11 and q = 03.
4. Explain RSA algorithm with example.
5. Explain Public Key Infrastructure (PKI) and its components in detail with diagram.
6. Explain digital signature with diagram.
7. Write steps involved in obtaining and verifying a digital certificate.
8. Describe applications of Public Key Cryptography in real-world systems.
9. Explain the role of Certificate Authority (CA) in secure communication.